

EMALIA HIGGINS

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EXPERIENCE

EXPEDIA GROUP — SOFTWARE DEVELOPMENT ENGINEER II — August 2023 - Present

- **Independently designed, built, and deployed a production rate limiting service for Escapia**, replacing a predecessor that suffered memory leaks and container crash loops; now handles ~323k requests/hour across a federated GraphQL API and a legacy SOAP reservation engine, reducing P99 latency 39.6%, P95 20.5%, and error rates 55.4% with zero post-deployment incidents. Built on Node.js/Hapi + AWS ElastiCache Valkey (cluster-mode + TLS, provisioned via Cerebro), with a custom Lua script executed via EVALSHA for atomic sliding-window rate limiting on multi-dimensional keys; owned K8s manifests, Helm charts, CI/CD, and the zero-downtime production cutover.
- **Migrated 20+ applications to containerized Kubernetes deployments across Kotlin/Java (Dropwizard), Node.js, and .NET 8 stacks** — authored K8s manifests and Helm charts, configured Spinnaker / Jenkins / GitHub Actions / Octopus Deploy pipelines with liveness/readiness/startup probes, and retired on-prem dependencies.
- **Collaborate cross-team on the continuous iteration of automated CI/CD pipelines (Spinnaker, Jenkins, GitHub Actions, Octopus Deploy)** to sustain service-owner feature velocity — reproducing and resolving production issues end-to-end, from log/trace triage in Splunk and Datadog through hotfix, pipeline update, and post-mortem.
- **Build and ship production-grade software with AI as a core part of the development workflow** — advanced Claude Code prompt engineering, skill/plugin authoring, agent orchestration, and MCP server integrations to deliver secure, optimized, high-quality code; **won Best Use of AI Award at the 2026 Expedia Group Global Hackathon** and published a VS Code extension (Claude Code Token Optimizer) that quantifies and reduces LLM token spend.
- **Defined availability and observability standards across the team toward the P&T 99.9% availability goal** — 30+ Splunk dashboards optimized via summary indexes, Datadog APM distributed tracing, SLO design, and PagerDuty/Slack integrated alerting;
- **On-call engineer in a 24/7 rotation for 2+ years**, triaging and resolving production incidents across 60+ services.
- **Achieved and maintained zero high-severity code and container vulnerabilities across 30+ applications**; authored the team's Tech Hygiene Score Guide and brought 5 tier-1 applications to a 100% hygiene score. Designed a double-scoped, encrypted, short-lived cross-trust-boundary token system (EdDSA-signed, RSA-OAEP-512-encrypted, JWS-in-JWE, 30-min TTL) so scope/property spoofing is blocked cryptographically.
- **Designed and shipped @expediagroup/vrbo-externalized-widgets — an internal React SDK and embeddable widget platform** that reduced time-to-ship-a-new-widget from 4.5 months to under a week and now powers 80+ property management agencies across React, Vue, Angular, and vanilla JS host pages.

INDEPENDENT PROJECTS

[TRO-NET.COM](#) — Founder / Solo Engineer

• Founded, designed, and built tro-net.com end to end — a scam-and-phishing email checker for older adults that returns a plain-English verdict on any forwarded email. Built the LLM classifier pipeline with code-level safety invariants (structurally incapable of surfacing an attacker phone number; runtime belt strips forbidden contact; hedged fallback), prompt-injection defenses on hostile inbound email, RFC-compliant outbound gating, Stripe billing, the 60+-first marketing site, and business ops (LLC, USPTO trademark). Stack: Next.js, TypeScript, Postgres/Neon + Drizzle, Inngest, Postmark, Stripe, Sentry.

GOOGLE EXPLORE CSR PROGRAM — January 2023 - May 2023

Selected for ML research; applied image feature extraction to Drosophila Genome Project data to train a Multi-Instance Multimodal Learning model for gene expression classification

INTERNSHIPS

EXPEDIA GROUP — Full Stack SDE Intern - Summer 2022

VENVEE, INC — Software Engineering Intern - August-December 2022

PATENTS

Co-Inventor — Personalized Vacation Listing Content Delivery System (U.S. Patent Pending)

VOLUNTEER WORK

BGOLDN Fresh Food Pantry (2yr), Kona Queens Hospital (1yr), Waimea Community Meal (3yrs)

AWARDS & RECOGNITION

- **Top 10 Finalist (3x), Expedia Group Global Hackathon, 2024 & 2025** — 2,000+ participants
- **Best Use of AI Award Winner**, Expedia Group Global Hackathon, 2026
- Selected as one of two engineers for Expedia Group's **SPX Technology Innovation V-Team** (Q3 2025)

EDUCATION

COLORADO SCHOOL OF MINES

BS, Computer Science

Student Athlete, Women's Soccer

SKILLS

AI Tooling & Prompt Engineering: Claude Code (advanced skills, plugins, agent orchestration, prompt engineering, cost/token optimization), MCP server integrations, Copilot, Cursor, Glean; published VS Code extension (Claude Code Token Optimizer)

Infrastructure & Orchestration:

Kubernetes, Helm, Docker, AWS (ElastiCache Valkey, S3, IAM, VPC), HashiCorp Vault, Spinnaker, Jenkins, GitHub Actions, Octopus Deploy

Distributed Systems & Storage: Redis / Valkey (cluster-mode, TLS), MongoDB, Apollo Federation (GraphQL), REST, SOAP, OpenAPI/Swagger contract-first design

Observability & Reliability: Datadog APM, Splunk (30+ production dashboards, summary indexes, alerts), SLI/SLO design, PagerDuty integration, 24/7 on-call incident response

Security: PKI / crypto-based auth (EdDSA, RSA-OAEP, JWS/JWE), least-privilege scoping, container & code vulnerability remediation at scale, cross-domain iframe sandboxing, secrets management, fail-open patterns

Languages (7+ years): Python, Node.js, TypeScript, Java, Kotlin, C#, SQL, Shell, GraphQL, Lua, React

Backend Frameworks: Hapi (Node), Dropwizard (Kotlin/Java), .NET 8

Testing: JUnit, Jest (with strict coverage thresholds), JMeter load testing, Postman, Bruno, SoapUI, Apollo Studio

CERTIFICATIONS

AWS Certified Developer Associate (2023)
AWS Certified Cloud Practitioner (2023)
Datadog APM and Distributed Tracing Fundamentals (2024)